RESEARCH ARTICLE

The Geology of Power: Weaponization of Critical Minerals as a New Domain of Geostrategic Coercion

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ABSTRACT

This paper examines the emergence of critical mineral supply chain weaponization as a novel domain of geostrategic coercion in the contemporary international system. Focusing on China's systematic deployment of export restrictions on rare earth elements, gallium, germanium, graphite, and antimony between 2023 and 2026, the study develops an original 'Mineral Coercion Escalation Model' (MCEM) that maps the progression from market dominance through selective restriction to full-spectrum weaponization. Through analysis of trade data, policy documents, and supply chain vulnerability assessments, this paper demonstrates that critical minerals have become the 21st century's most potent instrument of economic statecraft—surpassing energy resources in strategic leverage potential due to the extreme concentration of processing capacity and the absence of short-term substitutes. The research proposes a 'Geological Security Dilemma' framework, arguing that diversification efforts by the US, EU, and allied states are triggering a security spiral that intensifies rather than resolves supply chain vulnerabilities. The paper concludes with policy recommendations for a multilateral 'Critical Minerals Stability Pact' designed to prevent further escalation while ensuring equitable access to resources essential for the green and digital transitions.

Keywords: Critical Minerals, Economic Statecraft, Rare Earth Elements, Supply Chain Weaponization, Resource Geopolitics, China, Geostrategic Coercion

1. Introduction

The 21st century has witnessed the emergence of a new domain of geopolitical competition that reaches beneath the surface of the earth. Critical minerals—the lithium, cobalt, rare earth elements, gallium, germanium, and graphite essential for everything from electric vehicles to advanced semiconductors to precision-guided munitions—have become the most strategically consequential resources of the current era. Their importance rivals and may ultimately surpass that of petroleum in the 20th century, with one critical distinction: while oil reserves are distributed across multiple geographies, the processing of most critical minerals is concentrated to an extraordinary degree in a single state—China.

This paper argues that the weaponization of critical mineral supply chains represents a fundamentally new form of geostrategic coercion that the existing international relations literature has inadequately theorized. Unlike energy weaponization, which has been extensively studied since the 1973 oil crisis, mineral coercion operates through a different mechanism: control over processing and refining capacity rather than raw resource extraction. A nation may possess vast mineral deposits and still remain strategically vulnerable if it lacks the industrial capacity to transform raw ore into usable materials.

The period between 2023 and 2026 has seen a dramatic escalation in China's willingness to deploy mineral export restrictions as instruments of statecraft. Beginning with targeted controls on gallium and germanium effective August 1, 2023, Beijing has progressively expanded the scope and intensity of restrictions to encompass graphite, antimony, rare earth processing technologies, and most recently, comprehensive export licensing regimes covering materials critical to Western defense and technology supply chains. This escalation pattern demands systematic analysis.

2. Theoretical Framework: The Geological Security Dilemma

This paper introduces the concept of the 'Geological Security Dilemma' as an adaptation of the classical security dilemma framework for the domain of critical mineral supply chains. In its traditional formulation, the security dilemma describes a situation in which measures taken by one state to enhance its own security are perceived as threatening by other states, provoking countermeasures that leave all parties less secure than before.

The Geological Security Dilemma operates through an analogous mechanism in the resource domain. When the United States and its allies pursue supply chain diversification and mineral 'friend-shoring' to reduce dependence on Chinese processing, Beijing interprets these actions as strategic encirclement and responds with further export restrictions, which in turn validate and accelerate Western diversification efforts. The result is an escalatory spiral in which both sides' actions are individually rational but collectively destabilizing.

This dynamic is compounded by several features unique to the critical minerals domain. First, the extreme time lags involved in developing alternative mining and processing capacity—typically seven to fifteen years from discovery to production—mean that diversification efforts cannot provide near-term relief from supply vulnerabilities. Second, the environmental and social costs of mineral extraction and processing create political barriers to onshoring in democratic states. Third, the dual-use nature of most critical minerals—essential for both green energy transitions and military applications—creates internal policy tensions within importing states.

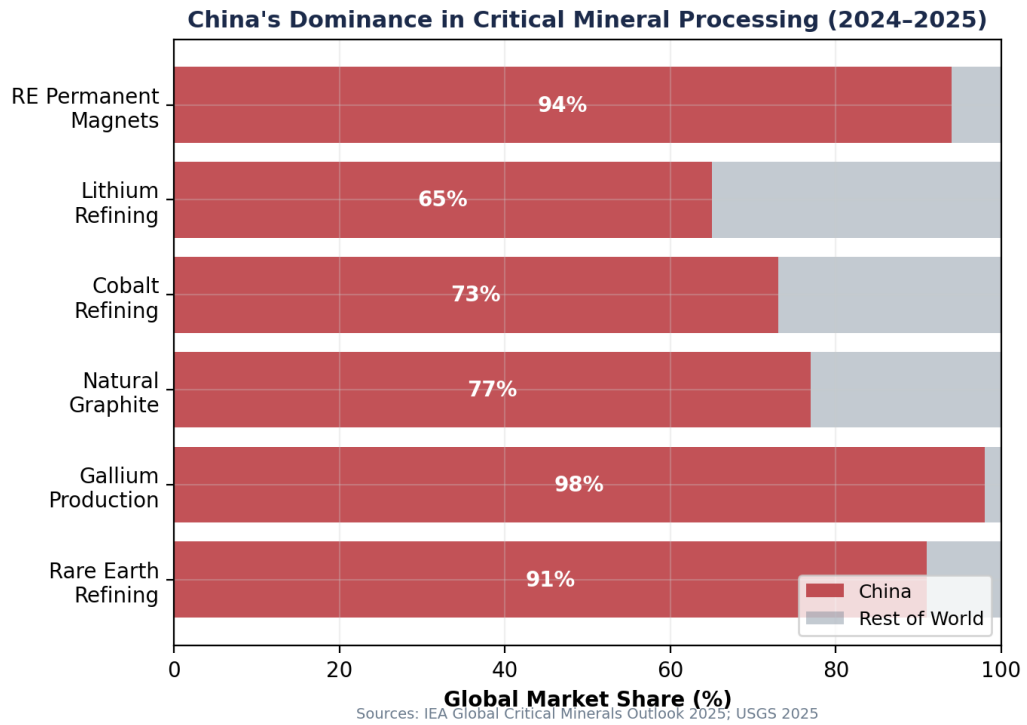

Figure 1: China's Dominance in Critical Mineral Processing — Global Market Share (2025)

3. The Escalation Trajectory: From Market Dominance to Weaponization

3.1 Phase I: Strategic Accumulation (2000–2020)

China's dominance in critical minerals did not emerge by accident. Beginning in the early 2000s, Beijing pursued a deliberate strategy of investing in mining operations across Africa, Latin America, and Central Asia while simultaneously building domestic refining capacity. State-subsidized processing facilities drove down global prices, rendering Western competitors uneconomic and creating the concentration of capacity that now provides strategic leverage. By 2020, China controlled approximately 90% of rare earth refining, 73% of cobalt refining, 65% of lithium refining, and 77% of natural graphite processing.

3.2 Phase II: Selective Restriction (2023–2024)

The turning point came in July 2023, when Beijing imposed export controls on gallium and germanium in direct response to US semiconductor export restrictions. These initial controls were carefully calibrated—targeting materials essential for chip manufacturing while leaving broader mineral flows undisturbed. The December 2023 extension to graphite controls, timed to coincide with intensified US-China technology tensions, signaled a willingness to expand the scope of mineral coercion. The antimony restrictions of 2024 targeted a material with direct military applications, including ammunition primers and infrared optics, marking the first explicit linkage between mineral controls and defense supply chains.

3.3 Phase III: Full-Spectrum Weaponization (2025–2026)

By late 2025, China had moved from selective restriction to comprehensive export licensing across the full spectrum of critical minerals. This phase was characterized by the extension of controls to rare earth processing technologies and equipment—targeting not just the materials themselves but the industrial capacity needed to produce them. The October 2025 export control overhaul demonstrated Beijing's willingness to weaponize its mineral dominance as a systemic instrument of statecraft rather than a targeted retaliatory tool.

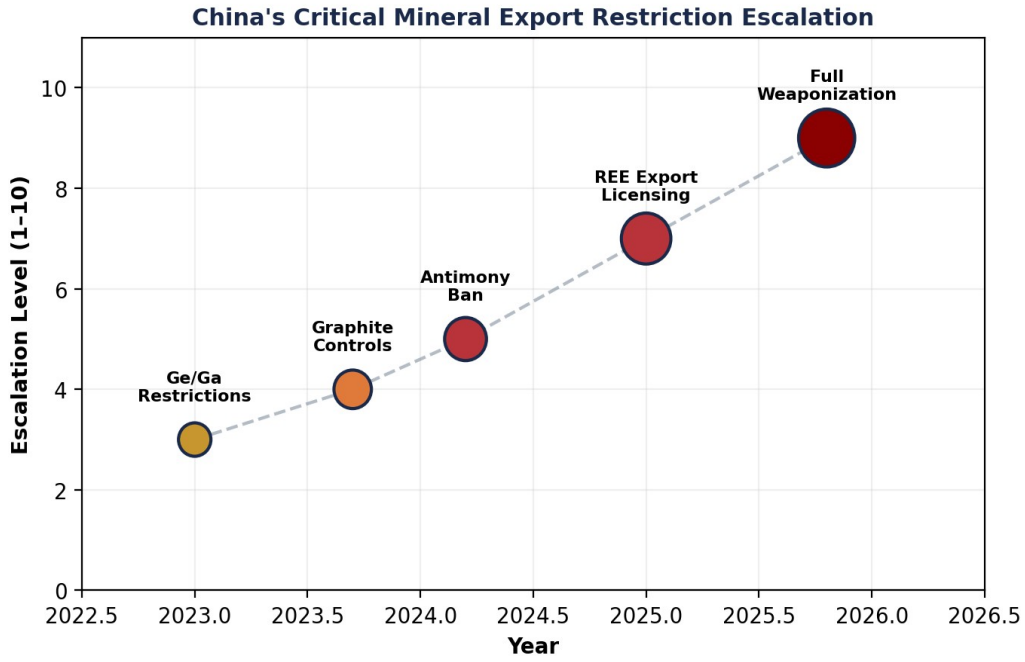

Figure 2: China's Critical Mineral Export Restriction Escalation Trajectory (2023–2026)

4. The Diversification Imperative and Its Contradictions

The Western response to China's mineral weaponization has centered on supply chain diversification—an agenda pursued through multiple channels including the US-led Minerals Security Partnership, the EU Critical Raw Materials Act, and bilateral agreements with resource-rich nations. However, this diversification imperative faces fundamental contradictions that limit its effectiveness in the near to medium term.

4.1 The Time Horizon Problem

New mining and processing operations require seven to fifteen years to move from exploration to commercial production. Even with accelerated permitting and regulatory frameworks, the fundamental geology, metallurgy, and engineering constraints of mineral extraction cannot be compressed. This creates a critical vulnerability window during which Western economies remain dependent on Chinese supply even as diplomatic relations deteriorate.

4.2 The Environmental Paradox

Critical mineral processing is among the most environmentally damaging industrial activities, involving toxic chemicals, massive water consumption, and significant carbon emissions. Democratic states face intense domestic opposition to the siting of processing facilities, creating a political obstacle to onshoring that authoritarian states do not face. The result is that the green transition—which drives much of the demand for critical minerals—is itself dependent on supply chains whose environmental costs undermine the transition's normative foundations.

4.3 The Development Dimension

Resource-rich developing nations, particularly in Africa, face the risk of being reduced to raw material suppliers in a new extractive relationship that replicates colonial patterns. The rush to secure alternative mineral sources has created a competitive dynamic among Western states, China, and Gulf sovereign wealth funds that offers short-term financial benefits to host nations but may entrench dependency on commodity exports rather than promoting industrial development and value-added processing capacity.

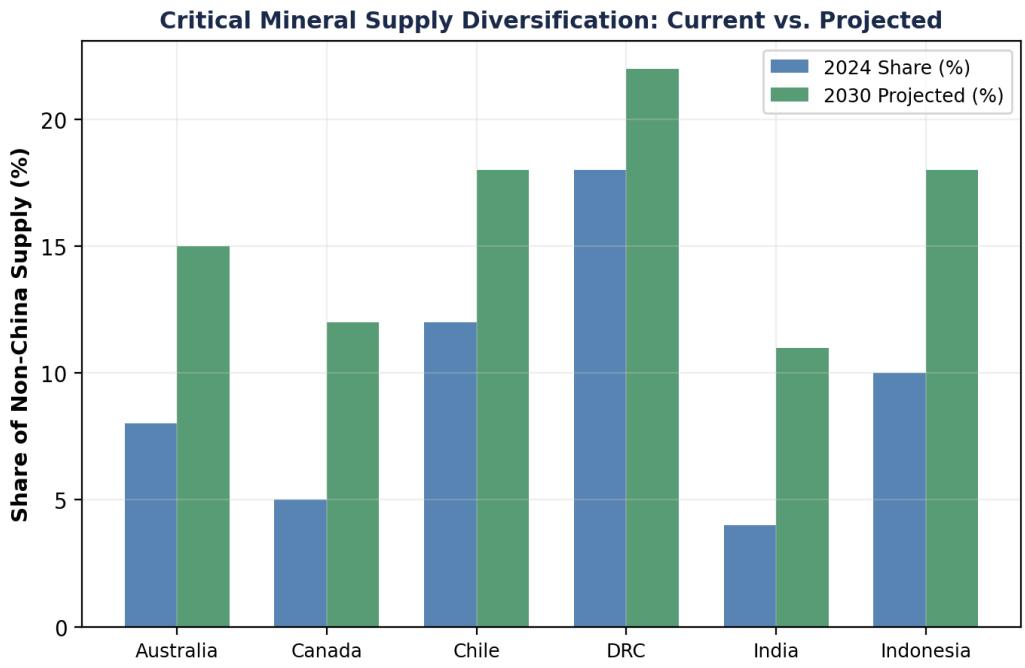

Figure 3: Critical Mineral Supply Diversification — Current vs. Projected Non-China Supply Shares

5. Strategic Implications

5.1 Defense Supply Chain Vulnerability

The defense implications of critical mineral dependency are particularly acute. Modern precision-guided munitions, advanced avionics, submarine propulsion systems, and electronic warfare equipment all depend on rare earth elements and specialty metals in which China dominates global supply. The United States and its NATO allies face the paradoxical situation of maintaining military alliances designed to deter China while depending on Chinese supply chains for the materials needed to manufacture the weapons systems that underpin deterrence. This contradiction represents one of the most significant strategic vulnerabilities in contemporary international security.

5.2 Alliance Cohesion Challenges

The scramble for alternative mineral sources is creating tensions within Western alliances. Competition between the United States and the European Union for access to Australian, Canadian, and African mineral resources has introduced friction into the transatlantic relationship. Bilateral mineral agreements—such as the US-DRC cobalt partnership and the EU-Chile lithium accord—risk creating a fragmented patchwork of preferential arrangements that undermines the multilateral coordination needed for effective diversification. The 'friend-shoring' concept, while strategically appealing, may prove economically inefficient and diplomatically divisive.

5.3 The Escalation Risk

Perhaps most concerning is the escalation potential inherent in the Geological Security Dilemma. If diversification efforts succeed in reducing Chinese market share, Beijing may respond by further

restricting remaining exports, hoarding strategic reserves, or deploying mineral leverage more aggressively in unrelated diplomatic contexts. Conversely, if diversification efforts fail, Western vulnerability increases, potentially encouraging more coercive Chinese behavior. Both trajectories carry significant risks for international stability.

6. Policy Recommendations: Toward a Critical Minerals Stability Pact

This paper proposes the establishment of a multilateral Critical Minerals Stability Pact (CMSP) designed to manage the Geological Security Dilemma through a combination of transparency, diversification, and restraint. The CMSP would incorporate four elements:

- Supply Transparency Mechanism: A multilateral reporting framework for critical mineral production, processing, trade, and reserves, modeled on the International Energy Agency's oil market transparency regime.
- Strategic Reserve Coordination: Coordinated strategic mineral reserves among allied nations, analogous to the IEA's coordinated petroleum reserve system, to buffer against supply disruptions and reduce the effectiveness of weaponized export restrictions.
- Processing Capacity Investment Fund: A multilateral investment vehicle to finance critical mineral processing capacity in diversified locations, with particular emphasis on value-added processing in resource-rich developing nations.
- Restraint Norms: A negotiated framework of mutual restraint on the weaponization of mineral supply chains, establishing graduated response protocols and dispute resolution mechanisms.

7. Conclusion

The weaponization of critical mineral supply chains represents one of the most significant and undertheorized developments in contemporary international security. China's systematic escalation from market dominance through selective restriction to full-spectrum weaponization has created a new domain of geostrategic coercion that demands urgent analytical and policy attention. The Geological Security Dilemma framework proposed in this paper illuminates the escalatory dynamics that make this domain particularly dangerous: actions taken to reduce vulnerability tend to provoke responses that increase it.

The critical minerals challenge is ultimately inseparable from the broader restructuring of the international order. The transition from fossil fuels to electrified energy systems, the digitalization of economies and militaries, and the intensification of great power competition all converge on the same set of geological resources. How the international community manages this convergence—whether through cooperative frameworks or competitive scrambles—will profoundly shape the character of the emerging international order. The Critical Minerals Stability Pact proposed in this paper represents one pathway toward managed competition; the alternative is a resource conflict that undermines both the green transition and international security.

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